Introduction to Mobile Robotics

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Robots !



Robots !



But also:

- Autonomous vehicles
- Drones
- Exoskeletons
- Vacuum cleaners
- etc

Robot ?

A machine that can

- **Sense:** Perceive through sensors (cameras, mic, etc)
- **Think:** Process info and decide (control algorithm)
- **Act:** Perform actions (moving, manipulate, interact)

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This course

1. Learn the basics principles using ROS2 (Robotic Operating System)
2. Implement simple behaviors in simulation, then on real robots
3. Realize a personal project involving a non-trivial autonomous (navigation) task

Our Robots



Pros

- Familiar, robust, extensible
- Various sensors and actuators
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- Various sensors and actuators
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Cons

- Lack the more advanced sensors (lidar, etc)
- don’t have many

Organization

- 11 sessions:
  - 4 sessions on basic concepts (1h30 lecture + 1h30 lab)
  - 2 advance labs + report
  - 5 sessions on personal projects
- Projects in groups of 4 max.
- No final exam, but intermediate report and defense/demo
- Pre-requisites:
  - Python, C/C++ (maybe optional)
  - Computers with native Ubuntu (UFR can lend a few laptops)
  - Dev tools: Git, Docker (will see basics in class)

Questions ?

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